

Mechanism hints v0.1 — cross-cutting CANDIDATE PATTERN
identified: many SM EW-scale observables have form
(substrate-primary-integer) · α = (substrate-integer)/N_max.
m_π, m_K, α_s, PMNS angles all fit. Pattern is CANDIDATE per
v1.4 brake; mechanism = multi-week.

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2026-05-30 Saturday 12:10 EDT (date-verified)

Mechanism hints — cross-cutting CANDIDATE pattern

Per v1.4 brake: this v0.1 is PATTERN OBSERVATION at CANDIDATE tier. No operational-criterion elevation.

0. The cross-cutting pattern

Today's CANDIDATE-arithmetic findings + existing BST observables show a consistent structural pattern at the EW scale:

SM observable $\approx$ (substrate-primary integer) · α = (substrate-integer) / N_max

where N_max = 137 = α^{-1} is the gauge-anchor.

1. Observables fitting this pattern

Observable	Substrate form	Substrate integer	Match
Schwinger a_e (existing T175)	$\alpha/(2\pi) = 1/(2\pi \cdot N_{\max})$	$1/(2\pi) \approx 0.159$	matches first-order
PMNS $\sin^2\theta_{13}$ (Elie Toy 3618)	$N_c/N_{\max} = 3/137$	3 (= N_c)	0.5%
PMNS $\sin^2\theta_{12}$ (Elie Toy 3618)	$\text{rank} \cdot N_c \cdot g / N_{\max} = 42/137$	42 (= rank · N_c · g)	0.1%
PMNS $\sin^2\theta_{23}$ (Elie Toy 3618)	$N_c \cdot n_C^2 / N_{\max} = 75/137$	75 (= N_c · n_C ²)	0.06%
**Cabibbo	V_us	**	$N_c^2 / (2^{N_c} \cdot n_C) = 9/40$
m_π / m_e (parallel batch 1)	rank · N_max = 274	274 = rank · N_max (= 2 · α^{-1})	0.3%
m_K / m_e (parallel batch 2)	g · N_max + rank · n_C = 969	g · N_max + small offset	0.3%
α_s(m_Z) (parallel batch 4)	$(N_c^2 + g) / N_{\max} = 16/137$	16 = N_c ² + g	0.94%
n_s (CMB) (existing)	1 – n_C/N_max	n_C correction	0.14%

Observable	Substrate form	Substrate integer	Match
Ω_m (matter density) (parallel batch 4)	$\text{rank} \cdot N_c \cdot g / N_{\text{max}} = 42/137$	42 (same as PMNS θ_{12} !)	2.7%

The pattern: SM observables at the EW / cosmological scale = (substrate-primary product) over $N_{\text{max}} = \alpha \cdot$ (substrate-product).

Notable: the integer 42 = $\text{rank} \cdot N_c \cdot g$ appears in BOTH PMNS $\sin^2 \theta_{12}$ AND Ω_m candidate. Suggestive of shared mechanism (the cosmological matter density and the solar mixing angle both involve the same substrate-primary product divided by the gauge anchor).

2. What this CANDIDATE pattern would mean

IF this pattern reflects a genuine mechanism (multi-week derivation needed), it would suggest: - SM gauge-anchored observables are uniformly = (substrate-integers) $\times \alpha$. - The substrate's natural quantization unit is $\alpha = 1/N_{\text{max}}$. - Different observables correspond to different (substrate-integer) numerators identifying the channel structure.

Cal #27 brake (internal): this pattern is OBSERVATIONAL. Many SM observables also fit different forms ($\sin^2 \theta_W = \text{rank}/N_c^2 = 2/9$ doesn't use N_{max}). So the pattern is partial, not universal. Treating it as a mechanism candidate requires multi-week derivation.

3. Honest disposition (v0.1, all CANDIDATE per v1.4)

- The CROSS-CUTTING PATTERN is suggestive.
- Multiple CANDIDATE arithmetic observables fit the (substrate-integer)/ N_{max} form.
- The pattern is NOT universal — $\sin^2 \theta_W$ and quark mass ratios don't fit.
- Promotion of any individual candidate, or of the pattern as a whole, to operational criterion requires:
 1. Mechanism derivation (kernel-integral / substrate-operator chain).
 2. Cal + Keeper cold-read PASS per audit-chain governance.
 3. Cal #99 META-distinction check (independence vs derived consequence).

Routed: → Keeper: cross-cutting pattern observation filed at CANDIDATE tier per v1.4 discipline. → Grace: pattern observation supports the “Reed-Solomon Ladder” structural reading (substrate-primaries cluster on EW-scale observables); cross-link to your Mendelev work. → Cal: cold-read this cross-pattern observation; mechanism gap for promotion. → me: continuing parallel work; this pattern observation = research signal not criterion.

— Lyra, mechanism hints v0.1. CROSS-CUTTING CANDIDATE PATTERN: SM EW-scale observables (PMNS angles, m_π , m_K , α_s , n_s , Ω_m , Schwinger) tend to have form (substrate-primary integer) $\cdot \alpha =$ (substrate-integer)/ N_{max} . NOT universal ($\sin^2 \theta_W$ and quark masses don't fit). Substrate-integer 42 = $\text{rank} \cdot N_c \cdot g$ appears in both PMNS θ_{12} AND Ω_m . CANDIDATE PATTERN per v1.4 brake; no criterion elevation; mechanism multi-week.